MA1522 Linear Algebra for Computing

Lesson 13: Orthogonality, and Orthogonal and Orthonormal Bases

Announcement: Online Quiz 3

- **Quiz 3 covers Chapter 3**, Sections 3.1–3.8.

- It opens this Saturday, **Saturday 3 October, 0:00**, and is due **Sunday 4 October, 23:59**.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson13` (e.g. `A1234567Z_lesson13.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 13**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

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Section 5.1 Orthogonality

Let $\mathbf{u} = (u_i)$ and $\mathbf{v} = (v_i)$ be vectors in $\mathbb{R}^n$.

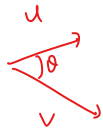
Inner product, norm and angle

$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n = \mathbf{u}^T \mathbf{v}, \qquad \|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = \sqrt{u_1^2 + \cdots + u_n^2}.$

In particular $\mathbf{u} \cdot \mathbf{u} \geq 0$, with equality if and only if $\mathbf{u} = \mathbf{0}$.

For $\mathbf{u}, \mathbf{v} \neq \mathbf{0}$, the **angle** between them is

$$\theta = \cos^{-1}\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|}\right), \qquad 0 \leq \theta \leq \pi.$$



$u \cdot v = \|u\| \cdot \|v\| \cos\theta$.

orthogonal $\Leftrightarrow \theta = 90^\circ \Leftrightarrow \cos\theta = 0 \Leftrightarrow u \cdot v = 0$.

Definition

1. $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** if $\mathbf{u} \cdot \mathbf{v} = 0$. For $\mathbf{u}, \mathbf{v} \neq \mathbf{0}$, this means the angle between them is $\theta = 90^\circ$.
2. A set $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is **orthogonal** if $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ for every $i \neq j$.
3. S is **orthonormal** if it is orthogonal and every $\mathbf{v}_i$ is a unit vector, $\|\mathbf{v}_i\| = 1$.

Example 1

True or false? Let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\} \subseteq \mathbb{R}^n$.

1. If $\mathbf{w} \cdot \mathbf{v}_i = 0$ for every i, then $\mathbf{w}$ is orthogonal to every vector in $\text{span}(S)$.
2. If S is orthonormal, then S is linearly independent.

1. Take $v \in \text{Span}(S)$. Then we can write. $v = c_1 v_1 + c_2 v_2 + \cdots + c_k v_k$.

$$w \cdot v = w \cdot (c_1 v_1 + \cdots + c_k v_k) = c_1 \underbrace{w \cdot v_1}_{0} + c_2 \underbrace{w \cdot v_2}_{0} + \cdots + c_k \underbrace{w \cdot v_k}_{0} = 0.$$

So w is orthogonal to $v \in \text{Span}(S)$.

2. Suppose $0 = c_1 v_1 + c_2 v_2 + \cdots + c_k v_k$

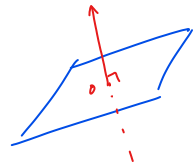
$$0 = 0 \cdot v_i = (c_1 v_1 + \cdots + c_k v_k) \cdot v_i = c_1 v_1 \cdot v_i + c_2 v_2 \cdot v_i + \cdots + c_k v_k \cdot v_i$$

c_i

$c_j v_j \cdot v_i = \begin{cases} c_j & \text{if } j = i \\ 0 & \text{if } j \neq i \end{cases}$

We conclude that $c_1 = c_2 = \cdots = c_k = 0$, so S is linearly independent.

Section 5.1 Orthogonal Complement



$\subseteq \mathbb{R}^3$.

Let V be a subspace of $\mathbb{R}^n$.

Definition

A vector $\mathbf{w}$ is **orthogonal to** V, written $\mathbf{w} \perp V$, if $\mathbf{w} \cdot \mathbf{v} = 0$ for every $\mathbf{v} \in V$. The **orthogonal complement** of V is

$$V^{\perp} = \{\, \mathbf{w} \in \mathbb{R}^n : \mathbf{w} \cdot \mathbf{v} = 0 \text{ for all } \mathbf{v} \in V \,\}.$$

Theorem

Let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be a spanning set for V.

1. $\mathbf{w} \perp V$ if and only if $\mathbf{w} \cdot \mathbf{u}_i = 0$ for all $i = 1, \ldots, k$.
2. Let $A = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_k \end{pmatrix}$. Then $V^{\perp} = \text{Null}(A^T)$.

$w \in \text{Null}(A^T)$

$\Leftrightarrow \quad A^T w = 0$

$$\begin{pmatrix} - & u_1^T & - \\ & \vdots & \\ - & u_k^T & - \end{pmatrix} w = \begin{pmatrix} u_1^T w \\ \vdots \\ u_k^T w \end{pmatrix} = \begin{pmatrix} u_1 \cdot w \\ \vdots \\ u_k \cdot w \end{pmatrix}$$

Question 1

$V^{\perp} = \text{Null}(A^T)$. $A = \begin{pmatrix} | & | & | \\ a_1 & a_2 & a_3 \\ | & | & | \end{pmatrix}$.

Let $V = \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\} \subseteq \mathbb{R}^5$, where

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \\ -1 \end{pmatrix}, \qquad \mathbf{a}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ -1 \\ 2 \end{pmatrix}, \qquad \mathbf{a}_3 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ 2 \\ -3 \end{pmatrix}.$$

Find a basis for $V^{\perp}$, and find $\dim(V)$ and $\dim(V^{\perp})$.

Question 1 — solution

$A = \begin{pmatrix} | & | & | \\ a_1 & a_2 & a_3 \\ | & | & | \end{pmatrix}$.

$V = \text{span}(a_1, a_2, a_3) = \text{Row}(A^T)$.

$V^\perp = \text{Null}(A^T)$

$= \begin{pmatrix} -a_1^T- \\ -a_2^T- \\ -a_3^T- \end{pmatrix}$

$A^T = \begin{pmatrix} 1 & 2 & 0 & 1 & -1 \\ 0 & 1 & 1 & -1 & 2 \\ 1 & 1 & -1 & 2 & -3 \end{pmatrix} \xrightarrow{\text{rref}} \begin{pmatrix} 1 & 0 & -2 & 3 & -5 \\ 0 & 1 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ → this gives basis of V.

$t_1 \quad t_2 \quad t_3$

sol. to $A^T x = 0 = \left\{ \begin{pmatrix} 2t_1 - 3t_2 + 5t_3 \\ -t_1 + t_2 - 2t_3 \\ t_1 \\ t_2 \\ t_3 \end{pmatrix} = t_1 \begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t_2 \begin{pmatrix} -3 \\ 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t_3 \begin{pmatrix} 5 \\ -2 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$.

$\dim(\text{span}(a_1, a_2, a_3))$

$\dim(V) \overset{?}{=} 2$.

$\Longrightarrow \left\{ \begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 5 \\ -2 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$ is a basis of $\text{Null}(A^T) = V^\perp$

$\dim(V^\perp) = 3$.

Section 5.2 Orthogonal and Orthonormal Bases

Definition

Let V be a subspace of $\mathbb{R}^n$. A set S is an **orthogonal basis** (resp. **orthonormal basis**) for V if S is a basis for V and S is an orthogonal (resp. orthonormal) set.

Theorem (Coordinates Relative to an Orthogonal Basis)

Let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be an orthogonal basis for V. Then for every $\mathbf{v} \in V$,

$$\mathbf{v} = \frac{\mathbf{v} \cdot \mathbf{u}_1}{\|\mathbf{u}_1\|^2}\mathbf{u}_1 + \cdots + \frac{\mathbf{v} \cdot \mathbf{u}_k}{\|\mathbf{u}_k\|^2}\mathbf{u}_k, \qquad \text{so} \qquad [\mathbf{v}]_S = \begin{pmatrix} \mathbf{v} \cdot \mathbf{u}_1 / \|\mathbf{u}_1\|^2 \\ \vdots \\ \mathbf{v} \cdot \mathbf{u}_k / \|\mathbf{u}_k\|^2 \end{pmatrix}.$$

If S is orthonormal, $[\mathbf{v}]_S = (\mathbf{v} \cdot \mathbf{u}_1, \ldots, \mathbf{v} \cdot \mathbf{u}_k)^T$.

$S = \{u_1, \ldots, u_k\}$.

Question 2

Suppose S is an orthogonal basis of V.

$V \Rightarrow v = c_1 u_1 + c_2 u_2 + \cdots + c_k u_k$.

To compute c_i, we inner product with u_i.

$v \cdot u_i = c_1 u_1 \cdot u_i + c_2 u_2 \cdot u_i + \cdots + c_k u_k \cdot u_i$

$= c_i u_i \cdot u_i$ ($u_j \cdot u_i = 0$ if $j \neq i$.)

$\Rightarrow c_i = \dfrac{v \cdot u_i}{u_i \cdot u_i} = \dfrac{v \cdot u_i}{\|u_i\|^2}$.

Let $V = \text{span}(S)$, where $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ and

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{u}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ -1 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 2 \\ -2 \\ 0 \\ 4 \end{pmatrix}.$$

(Hint: You can do all of the computation in one matrix multiplication)

Show that S is an orthogonal basis for V. Then, given that $\mathbf{v} \in V$, find $[\mathbf{v}]_S$ without solving a linear system.

Only need to show

- S is orthogonal → check $u_i \cdot u_j = 0$ if $i \neq j$.
  i.e. $\begin{cases} u_1 \cdot u_2 = 0 \\ u_2 \cdot u_3 = 0 \\ u_1 \cdot u_3 = 0 \end{cases}$.
- None of the member of S is 0.

Question 2 — solution

$$A^T = \begin{pmatrix} — u_1^T — \\ — u_2^T — \\ — u_3^T — \\ — v^T — \end{pmatrix}, \quad A = \begin{pmatrix} | & | & | & | \\ u_1 & u_2 & u_3 & v \\ | & | & | & | \end{pmatrix}$$

what are each entry?

$$= \begin{pmatrix} u_i^T u_j & u_i^T \cdot v \\ v^T \cdot u_j & v^T \cdot v \end{pmatrix} = \begin{pmatrix} u_i \cdot u_j & u_i \cdot v \\ v \cdot u_j & v \cdot v \end{pmatrix}$$

Challenge:

$S = \{u_1, \dots, u_k\}$.

The set S is orthogonal

$\Leftrightarrow A := \begin{pmatrix} | & & | \\ u_1 & \cdots & u_k \\ | & & | \end{pmatrix}$

A^TA is diagonal.

by MATLAB.

$$\begin{pmatrix} 4 & 0 & 0 & 8 \\ 0 & 4 & 0 & 4 \\ 0 & 0 & 4 & -4 \\ 8 & 4 & -4 & 24 \end{pmatrix}$$

a) $u_1 \cdot u_1 = 4$ $\quad u_1 \cdot u_2 = 0$ $\quad u_1 \cdot u_3 = 0$

$u_2 \cdot u_2 = 4$ $\quad u_2 \cdot u_3 = 0$

$u_3 \cdot u_3 = 4$.

$\Rightarrow \{u_1, u_2, u_3\}$ is orthogonal

b) Thm gives $v = \frac{v \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{v \cdot u_2}{u_2 \cdot u_2} u_2 + \frac{v \cdot u_3}{u_3 \cdot u_3} u_3$ ($v \cdot u_1 = 8$, $u_1 \cdot u_1 = 4$, $v \cdot u_2 = 4$, $u_2 \cdot u_2 = 4$, $v \cdot u_3 = -4$, $u_3 \cdot u_3 = 4$)

$= 2u_1 + 1u_2 - u_3$. $\quad [v]_S = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$.

Challenge 1

Let $V = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be as in Question 2. For $\mathbf{w} \in \mathbb{R}^4$, let $c_i = \mathbf{u}_i \cdot \mathbf{w}$ for $i = 1, 2, 3$, and set

$$\mathbf{w}' = \mathbf{w} - \left(\frac{c_1}{\|\mathbf{u}_1\|^2}\mathbf{u}_1 + \frac{c_2}{\|\mathbf{u}_2\|^2}\mathbf{u}_2 + \frac{c_3}{\|\mathbf{u}_3\|^2}\mathbf{u}_3 \right).$$

1. Show that $\mathbf{w}' \in V^{\perp}$.
2. Show that $\mathbf{w} \in V$ if and only if $\mathbf{w}' = \mathbf{0}$.

Section 5.2 Orthogonal Matrices

Let $Q = (\mathbf{u}_1\ \mathbf{u}_2\ \cdots\ \mathbf{u}_k)$ be an $n \times k$ matrix. The (i,j)-entry of Q^TQ is $\mathbf{u}_i \cdot \mathbf{u}_j$, so $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ is orthonormal if and only if $Q^TQ = I_k$.

Definition

A square matrix A of order n is **orthogonal** if $A^T = A^{-1}$, equivalently, $A^TA = I_n = AA^T$.

Theorem

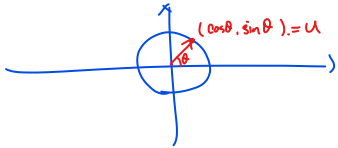
Let A be a square matrix of order n. The following are equivalent.

(i) A is an orthogonal matrix.

(ii) The columns of A form an orthonormal basis for $\mathbb{R}^n$.

(iii) The rows of A form an orthonormal basis for $\mathbb{R}^n$.

Question 3



You may assume that $x^2 + y^2 = 1$ if and only if $x = \cos\theta$ and $y = \sin\theta$ for some θ.

1. Let $\mathbf{u} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$. Suppose that the 2×2 matrix $(\mathbf{u}\ \mathbf{v})$ is orthogonal. Find all possible $\mathbf{v}$.

$A = \begin{pmatrix} \cos\theta & x \\ \sin\theta & y \end{pmatrix}$ solve x, y s.t. $A^T A = Id$.

2. Let $\mathbf{u} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. Suppose that the 3×3 matrix $(\mathbf{u}\ \mathbf{v}\ \mathbf{w})$ is orthogonal. Find all possible $\mathbf{v}$ and $\mathbf{w}$.

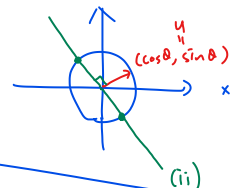
Question 3 — solution

a) $A = \begin{pmatrix} \cos\theta & x \\ \sin\theta & y \end{pmatrix}$. Solve $\begin{pmatrix} x \\ y \end{pmatrix}$ s.t. $A^TA = Id$.

$$\begin{pmatrix} \cos\theta & \sin\theta \\ x & y \end{pmatrix}\begin{pmatrix} \cos\theta & x \\ \sin\theta & y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

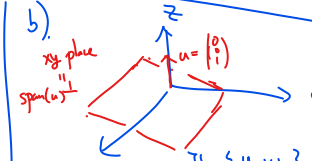
A is orthogonal $\Leftrightarrow$ its columns form an orthonormal basis of $\mathbb{R}^2$.

$\overset{(i)}{\Leftrightarrow} \left\| \begin{pmatrix} x \\ y \end{pmatrix} \right\|^2 = 1$ and $\overset{(ii)}{\begin{pmatrix} x \\ y \end{pmatrix}}$ is orthogonal to $\begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$.



So $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos(\theta \pm 90^\circ) \\ \sin(\theta \pm 90^\circ) \end{pmatrix} = \begin{pmatrix} \sin\theta \\ -\cos\theta \end{pmatrix}$ or $\begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$.

b)

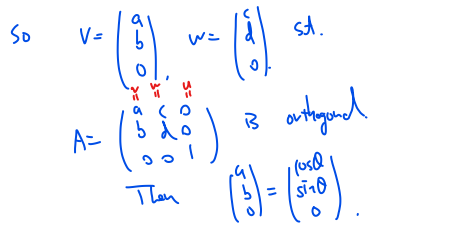


If $\{u, v, w\}$ forms an orthonormal basis, then $v, w \in \text{span}(u)^\perp$.

Challenge 2

Let A and B be orthogonal matrices of order n.

1. Show that $\det(A) = 1$ or $\det(A) = -1$.
2. Show that AB is an orthogonal matrix.
3. Show that A^{-1} is an orthogonal matrix.



Next lesson

Next lesson (**Fri 2 Oct**, Sections 5.2 & 5.3), I will focus on

1. the definition of orthogonal projection, and how to compute it;
2. how to perform Gram-Schmidt Orthogonalization to transform a basis into an orthonormal basis, on an explicit example.

Please watch the video clips beforehand.